

Semester	B.E. Semester VII – Computer Engineering
Subject	Devops
Subject In-charge	Dr. Snehal Andhare

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Roll Number	23102B0008
Grade and Subject Teacher's Signature	

Practical Number	02
Practical Title	To understand Version Control System / Source Code Management, install git and create a GitHub account. To Perform various GIT operations on local and Remote repositories using GIT Cheat-Sheet.
Objectives (Skill Set/ Knowledge Tested/ Imparted)	• To get the idea about how github and git desktop works. • How to use gitt bash terminal • Understanding basic commands of git • Getting to know about versioning systems • How git works internally.
Description	We tried different devops commands for github via installing github desktop and creating local chngs and then pushing it to the remote branch in main

Program &
Output

```
23102B0006@VSIT-M514-302 MINGW64 ~
$ cd devops

23102B0006@VSIT-M514-302 MINGW64 ~/devops
$ pwd
/c/Users/23102B0006/devops

23102B0006@VSIT-M514-302 MINGW64 ~/devops
$ touch dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops
$ ls
dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops
$ notepad dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops
$ ls
dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops
$ git init
Initialized empty Git repository in C:/Users/23102B0006/devops/.git/

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ ls -m
dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    dev.html

nothing added to commit but untracked files present (use "git add" to track)

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git add dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git status
On branch master
```

```
23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ git branch
* main

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ git branch branchname

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ git branch
branchname
* main

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ git checkout branchname
M      dev.html
Switched to branch 'branchname'

23102B0006@VSIT-M514-302 MINGW64 ~/devops (branchname)
$ touch new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (branchname)
$ notepad new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (branchname)
$ ls
dev.html  new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (branchname)
$ git add new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (branchname)
$ git commit -m "initial commit b"
[branchname 10ef095] initial commit b
1 file changed, 14 insertions(+)
create mode 100644 new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (branchname)
$ git checkout main
M      dev.html
Switched to branch 'main'

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ ls
dev.html
```

```
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git add .

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git diff --staged
diff --git a/dev.html b/dev.html
new file mode 100644
index 0000000..b677a7a
--- /dev/null
+++ b/dev.html
@@ -0,0 +1,14 @@
+<!DOCTYPE html>
+<html lang="en">
+<head>
+  <meta charset="UTF-8">
+  <meta name="viewport" content="width=device-width, initial-scale=1.0">
+  <title>My Simple Web Page</title>
+</head>
+<body>
+  <h1>Welcome to My Website</h1>
+  <p>This is a simple paragraph of text on your brand-new HTML page.</p>
+  <a href="https://wikipedia.org" target="_blank">Visit Wikipedia</a>
+</body>
+</html>

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git commit -m "initial commit"
[master (root-commit) 01f9293] initial commit
1 file changed, 14 insertions(+)
create mode 100644 dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ notepad dev.html
bash: notepad: command not found

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ notepad dev.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git status
```

```
23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   dev.html

no changes added to commit (use "git add" and/or "git commit -a")

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git diff
diff --git a/dev.html b/dev.html
index b677a7a..4462cc2 100644
--- a/dev.html
+++ b/dev.html
@@ -9,6 +9,6 @@
        <h1>Welcome to My Website</h1>
        <p>This is a simple paragraph of text on your brand-new HTML page.</p>
-       <a href="https://wikipedia.org" target="_blank">Visit Wikipedia</a>
+       <p>Changed the FILE</p>
</body>
</html>

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git log
commit 01f9293f1e3f822d4f855656dbf7ef579180ab4e (HEAD -> master)
Author: [adi chavan] <[adityachavan1357@gmail.com]>
Date:   Tue Aug 4 12:13:52 2026 +0530

    initial commit

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git branch]
git: 'branch]' is not a git command. See 'git --help'.

The most similar command is
    branch

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git branch
* master

23102B0006@VSIT-M514-302 MINGW64 ~/devops (master)
$ git branch -M main

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ git branch
    branchname
* main

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ git merge branchname
Updating 01f9293..10ef095
Fast-forward
 new.html | 14 ++++++++
 1 file changed, 14 insertions(+)
 create mode 100644 new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ ls
dev.html  new.html

23102B0006@VSIT-M514-302 MINGW64 ~/devops (main)
$ |
```

Conclusion:

Got to know how to work with git terminal, github desktop and github as a version controlling system

	Worked with how to use and know the versioning of same files in different branches
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